

Sub phase-2 solution requirement

Date	28-09-2026
Team ID	SWTID-2026-7624
Project name	Script-controlled ACL
Maximum marks	4 marks

3. Solution Requirements

Solution requirements define the functions and quality characteristics required to implement the Script-Controlled ACL system successfully.

A. Functional Requirements

1. User Authentication

The system shall allow authorized users to log in to the ServiceNow platform using valid credentials.

2. Role Management

The system shall identify and manage users based on their assigned roles.

3. Field-Based Access Restriction

The system shall restrict record access based on the value of a specific field such as Branch.

4. Read Access

Users with the bb1 role shall be allowed to view records according to the defined branch-based access condition.

5. Create Access

Users assigned with the bb2 role shall be provided with permission to create new records.

6. Write Access

Users assigned with the bb3 role shall be allowed to modify or edit existing records.

7. Delete Access

Users assigned with the bb4 role shall be allowed to delete records.

8. Administrative Override

Users with the admin role shall have unrestricted access to all records.

9. Access Denial

Users without the required permissions shall be prevented from accessing restricted records.

10. Access Validation

The system shall allow administrators to test access permissions using ServiceNow's impersonation feature.

B. Non-Functional Requirements

1. Security

The system must provide secure and controlled access to organizational records.

2. Reliability

The access-control mechanism should consistently apply the defined rules for every access request.

3. Performance

The ACL evaluation should be performed efficiently without causing unnecessary delay while accessing records.

4. Scalability

The system should support additional users, roles, branches, and access conditions when required.

5. Maintainability

The ACL scripts and role-based permissions should be easy to maintain and modify.

6. Usability

Authorized users should be able to access the records relevant to their responsibilities without unnecessary complexity.

7. Flexibility

The system should support dynamic access decisions instead of depending only on multiple static ACL rules.

8. Availability

The access-control functionality should be available whenever authorized users access the ServiceNow application.

3. Solution Requirements

Solution requirements define the functions and quality characteristics required to implement the Script-Controlled ACL system successfully.

A. Functional Requirements

No.	Requirement	
1	User Authentication	Allow authorized users to log in to ServiceNow.
2	Role Management	Identify and manage users based on their roles.
3	Field-Based Restriction	Restrict access based on the value of a specific field (Branch).
4	Read Access	bb1 role can view records based on branch condition.
5	Create Access	bb2 role can create new records.
6	Write Access	bb3 role can modify or edit existing records.
7	Delete Access	bb4 role can delete records.
8	Administrative Override	admin role has unrestricted access.
9	Access Denial	Block users without required permissions.
10	Access Validation	Allow administrators to test access using impersonation.

B. Non-Functional Requirements

No.	Requirement	
1	Security	Provide secure and controlled access to organizational records.
2	Reliability	Apply access rules consistently for every request.
3	Performance	ACL evaluation should be efficient without delay.
4	Scalability	Support additional users, roles, branches and conditions.
5	Maintainability	ACL scripts and role permissions should be easy to maintain.
6	Usability	Allow authorized users to access relevant records easily.
7	Flexibility	Support dynamic access decisions (not only static ACL rules).
8	Availability	Access-control functionality should be available whenever needed.